

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the static characteristics of instrument.
- Q.24 Explain in detail various sources of errors in measurement.
- Q.25 Explain and draw building blocks of generalized measurement system.

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Roll No.

1st Sem / Instrumentation & Control engg.

Subject : Fundamentals of Instrumentation Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following error is caused by poor calibration of the instrument?
- a) Random error b) Gross error
- c) Systematic error d) Precision error
- Q.2 Transducers which used external power are called
- a) Passive transducer b) Active transducer
- c) Both A & B d) None of these
- Q.3 Fluorescent is a
- a) Recording device b) Display device
- c) Calibration device d) Sensing device

- Q.4 Which of the following represent active transducer?
- a) Strain gauge b) Thermistor
- c) Potentiometer d) Thermocouple
- Q.5 In a measurement, what is the term used to specify the closeness of two or more measurement?
- a) Precision b) Accuracy
- c) Fidelity d) Threshold
- Q.6 A measuring system consists of
- a) Sensors
- b) variable conversion elements
- c) Signal processing elements
- d) All of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Expand LED
- Q.8 The switching time of LED is of the order of_____.
- Q.9 The ratio of maximum displacement deviation to full scale deviation of the instrument is called_____.

- Q.10 Torque/weight ratio of an instrument represents _____
- Q.11 All sensors are transducers.(T/F)
- Q.12 During a measurement, for a measure value “B”, absolute error is obtained as “A”, what will be the relative error of measurement?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the selection criteria of instruments.
- Q.14 What is relative limiting error?
- Q.15 What is natural frequency?
- Q.16 Explain the characteristics of LCD.
- Q.17 What is the primary sensing element in measurement system?
- Q.18 What is the difference between sensor and transducer.
- Q.19 What is normal distribution of errors?
- Q.20 What is classification of instrumentation system?
- Q.21 Give 3 examples of each indicating and recording instruments.
- Q.22 Explain Magnetic tape recorder with diagram.